Instructions:

- Clearly explain all your answers.
- For each True/False question, state your answer and provide a brief justification.
- Make necessary assumptions and clearly state all the assumptions.

Submission Date: August 3, 2025, 5:00 PM.

Problem 1: Let $\mathbb{R}$ be the set of real numbers, U be a subspace of $\mathbb{R}^3$, and $M \in \mathbb{R}^{3\times 3}$ be the matrix corresponding to the projection onto the subspace U. Which of the following statements is/are TRUE?

1. If U is a 1-dimensional subspace of $\mathbb{R}^3$, then the null space of M is a 1-dimensional subspace.
2. If U is a 2-dimensional subspace of $\mathbb{R}^3$, then the null space of M is a 1-dimensional subspace.
3. $M^2 = M$
4. $M^3 = M$

Problem 2: Compute the projection matrices onto the lines through $a_1 = (-1, 2, 2)$ and $a_2 = (2, 2, -1)$. Multiply those projection matrices and explain why their product P_1P_2 is what it is.

Problem 3: Project $b = (1, 0, 0)$ onto the lines through a_1 and a_2 in Problem 2 and also onto $a_3 = (2, -1, 2)$. Add up the three projections $p_1 + p_2 + p_3$.

Problem 4: Continuing Problems 2–3, find the projection matrix P_3 onto $a_3 = (2, -1, 2)$. Verify that $P_1 + P_2 + P_3 = I$. The basis a_1, a_2, a_3 is orthogonal!

Problem 5: Let W_1, W_2, W_3 be the subspaces of $\mathbb{R}^3$ given by

$$W_1 = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}, \quad W_2 = \{(x, y, z) \in \mathbb{R}^3 : x = z\},$$
$$W_3 = \{(0, 0, z) \in \mathbb{R}^3 : z \in \mathbb{R}\}.$$

Prove that $\mathbb{R}^3 = W_1 + W_3$ and $\mathbb{R}^3 = W_2 + W_3$. Which of these sums are direct?

Problem 6. Write the complete solution as x_p plus any multiple of s in the nullspace:

$$\begin{aligned} x + 3y + 3z &= 1, \\ 2x + 6y + 9z &= 5, \\ -x - 3y + 3z &= 5. \end{aligned}$$

Problem 7: Find bases and dimensions for the four fundamental subspaces associated with the matrices.

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 8 \end{bmatrix}.$$

Problem 8: If V is the subspace spanned by (1, 1, 1) and (2, 1, 0), find a matrix A that has V as its row space. Find a matrix B that has V as its nullspace. Multiply AB.

Problem 9: If W_1 and W_2 are subspaces of $V(F)$, then prove that $W_1 \cup W_2$ is a subspace of $V(F)$ if and only if either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Problem 10. prove that:
a) intersection of 2 subspace of vector space is subspace.
b) union of subspaces may or may not be subspace.